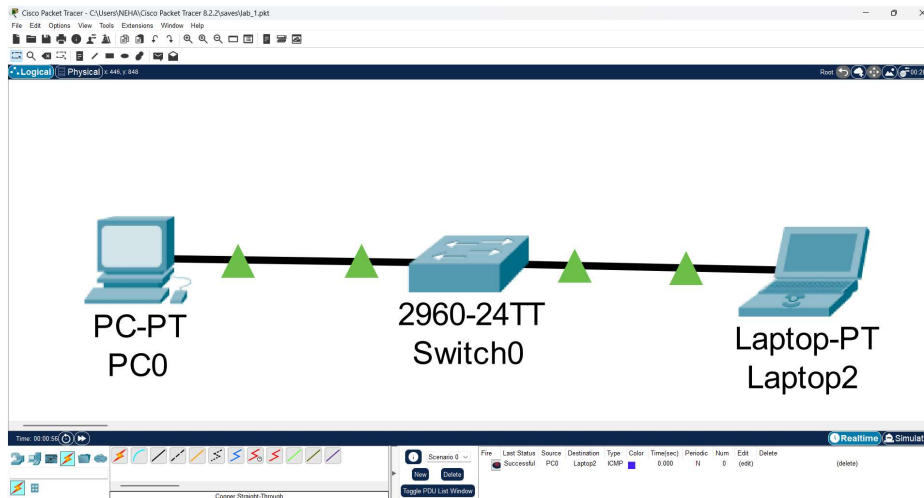


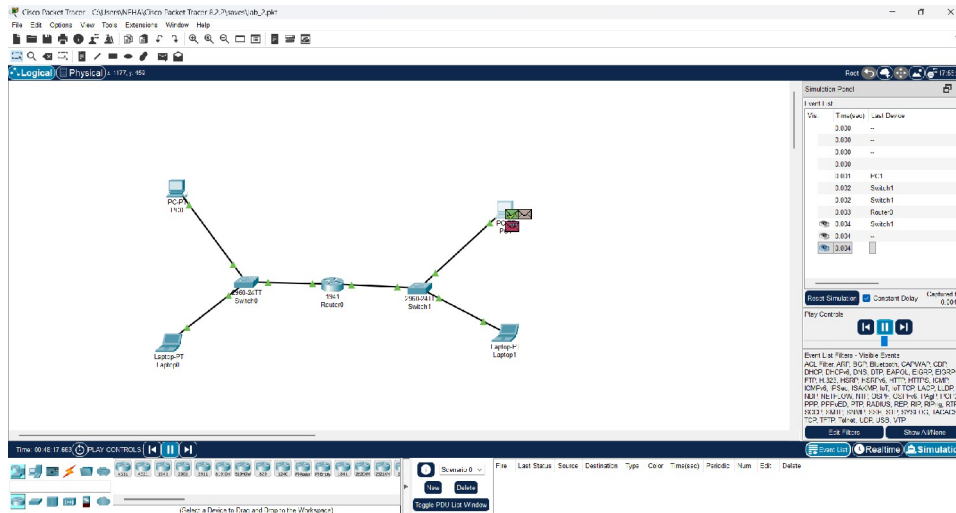
U24ai05 Lab-2

13 January 2026 12:07

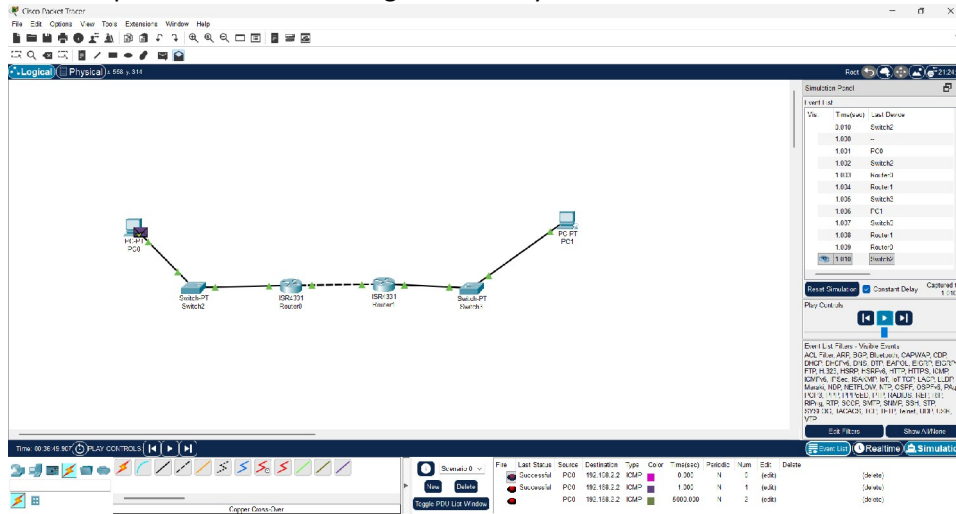
1. Construct a virtual LAN in the Cisco packet tracer as shown below, and check where the data packets are transferring from one system to another.

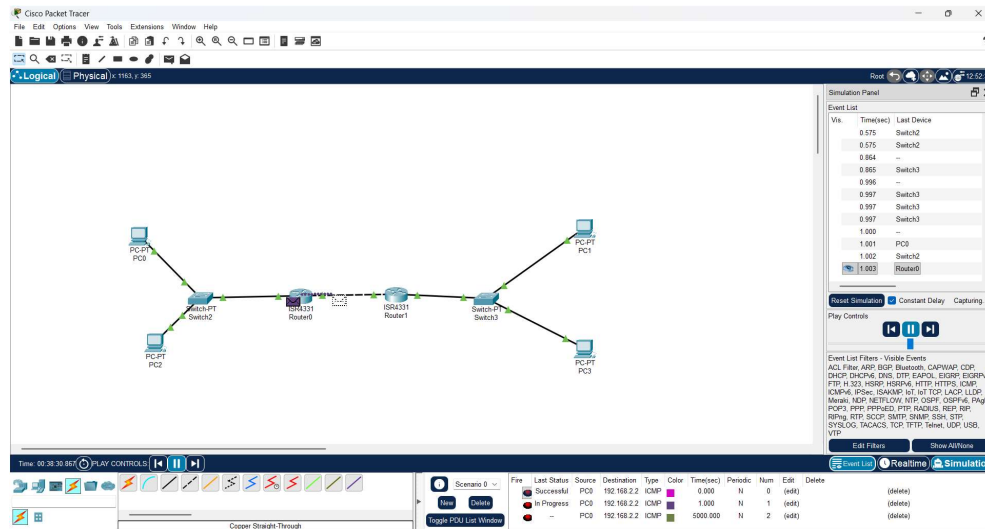


2. Construct a virtual MAN in the Cisco packet tracer, as shown below, and check where the data packets transfer from one system to another.



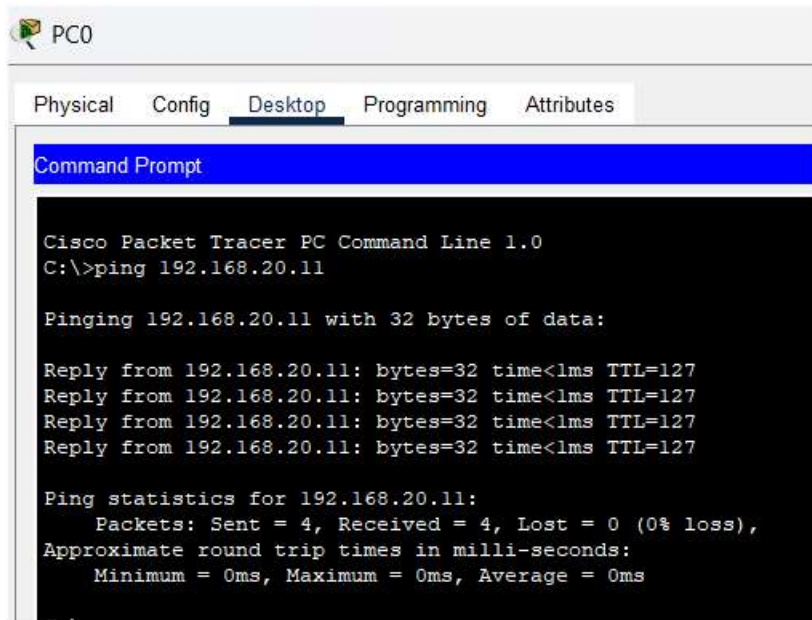
3. Construct a virtual WAN in the Cisco packet tracer as shown below, and check where the data packets are transferring from one system to another.





Q-4 In Cisco Packet Tracer, design a campus network with two buildings (Building-A and Building-B). Each building should have at least two IP subnets (e.g., Subnet 10–Students and Subnet 20–Faculty) implemented using switches. Connect both buildings via routers (using either subinterfaces or inter-router links), assign suitable IP subnets to each group, configure links between switches and routers, set up inter-subnet routing so that only Faculty subnets of both buildings can communicate with each other while Student subnets remain isolated, and verify this behavior using ping and simulation mode to observe packet flow

Student to teacher



The screenshot shows the 'PC0' window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The text in the command prompt is as follows:

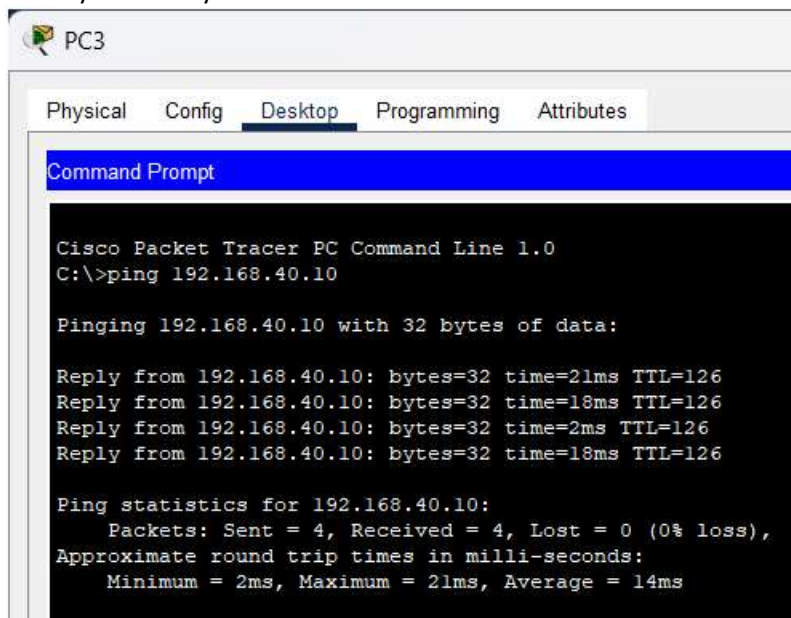
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Faculty to Faculty



The screenshot shows the 'PC3' window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The text in the command prompt is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.40.10: bytes=32 time=21ms TTL=126
Reply from 192.168.40.10: bytes=32 time=18ms TTL=126
Reply from 192.168.40.10: bytes=32 time=2ms TTL=126
Reply from 192.168.40.10: bytes=32 time=18ms TTL=126

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 21ms, Average = 14ms
```

Student to Student NO communication possible

```

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

Router to Router

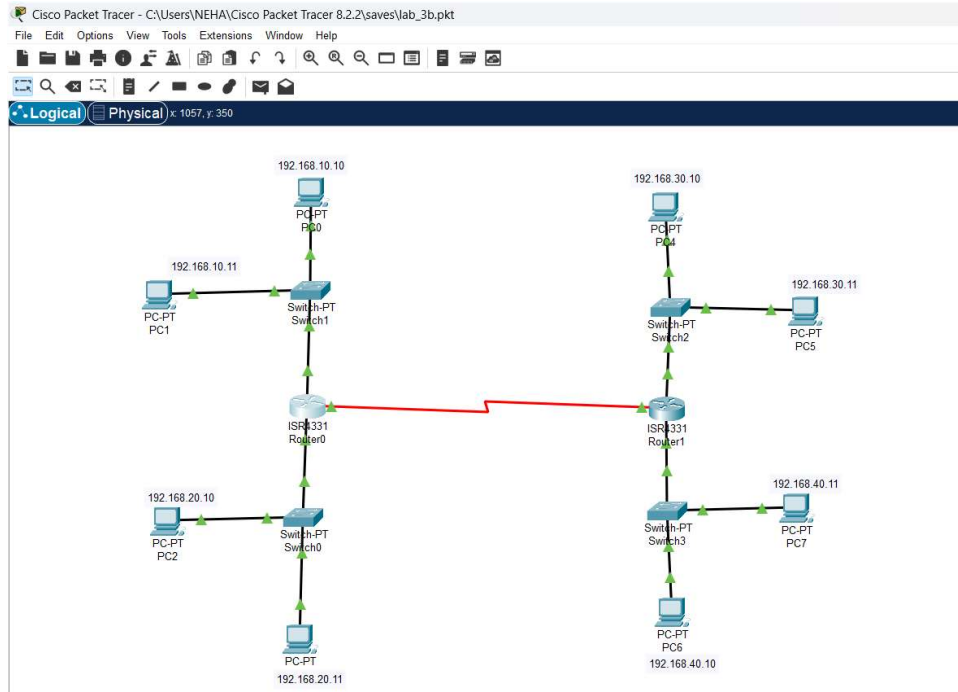
```

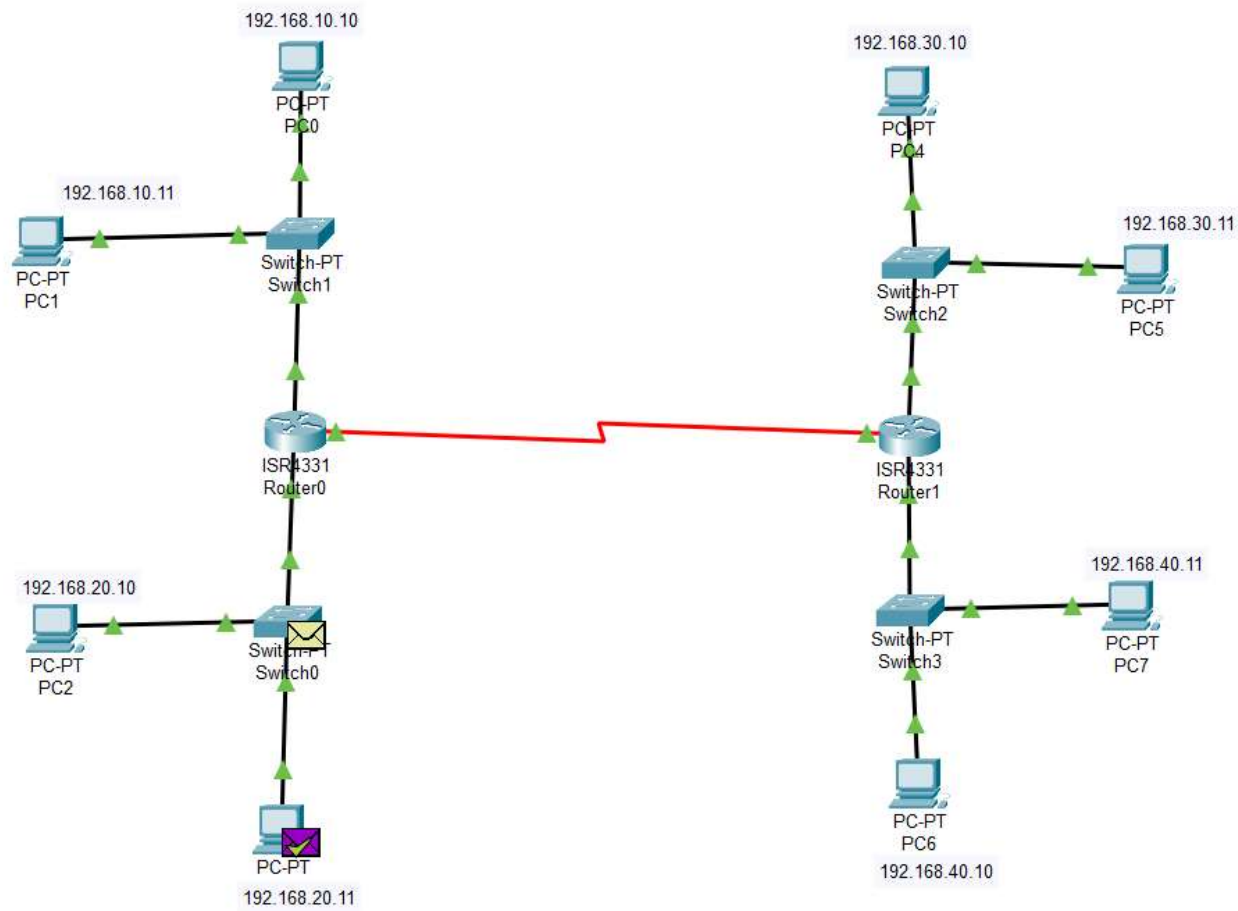
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 6/16/49 ms

```





Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC2	PC0	ICMP	Yellow	0.000	N	1	(edit)	(delete)
	Failed	PC1	PC6	ICMP	Purple	0.000	N	2	(edit)	(delete)
	Successful	PC3	PC6	ICMP	Brown	0.000	N	3	(edit)	(delete)
	Successful	PC3	PC7	ICMP	Purple	0.000	N	4	(edit)	(delete)